

Binghang Lu

West Lafayette, IN

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Research Interests

Scientific Machine Learning, Large Language Models, Continual Learning, Generative AI

Education

Purdue University, West Lafayette, IN *Jan 2025 – May 2028*

Ph.D. in Electrical and Computer Engineering

Focus: Generative AI, Large Language Models, Scientific Machine Learning

Purdue University, West Lafayette, IN *Jan 2025 – Dec 2026*

M.S. in Electronic and Computer Engineering

Focus: Generative AI, Large Language Models, Scientific Machine Learning

Purdue University, West Lafayette, IN *Aug 2021 – Dec 2024*

B.S. in Computer Science and Mathematics

Honors: Dean's List, Semester Honors

Honors and Awards

- Featured Research Interview, National Academy of Inventors Convention 2026
- Institute for Computational and Experimental Research in Mathematics (ICERM) Travel Grant, Brown University Apr 2026
- NSF Discover Allocation, Principal Investigator, awarded \$750,000 in computing resources 2025
- Institute for Mathematical and Statistical Innovation (IMSI) Travel Award 2025
- Spira Research Fellowship, College of Science, Purdue University 2024
- Dean's List and Semester Honors, Purdue University 2024
- Thomas Arai Scholarship, Department of Mathematics, Purdue University 2022

Research Experience

Harvard University *Aug 2025 – Present*

Visiting Research Assistant

- Conduct research on optimization methods for continual learning and large language models.
- Proposed spectral-guided optimization ideas for efficient training in scientific machine learning and LLM settings.
- Investigated constrained and projection-based optimization methods to mitigate catastrophic forgetting.

Purdue University, Lin's Lab *Jan 2022 – Present*

Research Assistant

- Conduct research in scientific machine learning, physics-informed neural networks, neural operators, optimization, and AI systems.

- Developed optimization frameworks for PINNs and operator learning, including multi-objective and evolutionary approaches.
- Worked on ensemble Kalman filter enhanced PINNs, time-fractional PDE operator learning, and replica-exchange-based optimization.

Carnegie Mellon University, Pittsburgh, PA
Machine Learning Summer Intern

May 2020 – Aug 2020

- Developed and fine-tuned a YOLOv5 computer vision model for mask-wearing recognition.
- Contributed to a publication on object detection for mask-wearing recognition.

Industry Experience

AT&T Labs
Sr. Associate Research Intern Technical II, Generative AI

Jun 2026 – Aug 2026

- Worked on machine learning methods for predicting user performance under varying network conditions.
- Built a **multi-agent LLM system** with LangGraph/LangChain and **vLLM**, adding request batching, KV-cache optimization, and agent orchestration to reduce **p95 latency by 40%**.
- Designed scalable **RAG pipelines** with hybrid retrieval, intelligent chunking, and vector database indexing for enterprise knowledge-intensive reasoning tasks.
- Developed **short- and long-term agent memory** with retrieval caching, reducing repeated inference cost and improving workflow-level context retention.
- Built **agent-based monitoring** with MCP/A2A protocols for telemetry collection, anomaly detection, and automated remediation in production workflows.

Microsoft
Research Collaborator

Aug 2025 – Present

- Collaborating with Dr. Xiaomin Li to formulate and develop **Muon-OGD**, a novel spectral-norm-regularized constrained optimization framework to mitigate catastrophic forgetting during Large Language Model (LLM) post-training.
- Architected a low-rank variant of the Muon optimizer utilizing Randomized Singular Value Decomposition (SVD) and Power Iterations to improve computational efficiency and gradient approximation.
- Scaled the proposed optimization algorithm across a distributed 100+ GPU cluster, empirically demonstrating a 10% training acceleration on NanoGPT models compared to standard Muon baselines.
- Co-authoring a manuscript detailing the algorithm's convergence properties and empirical performance for submission to **NeurIPS 2026**.

Wavelogix, Inc., West Lafayette, IN
Machine Learning Engineer Intern

May 2025 – Oct 2025

- Developed multimodal recommendation and generative AI models for real-world sensor and user interaction data.

- Built scalable Python and MongoDB data pipelines for large-scale IoT sensor processing.
- Improved preprocessing efficiency and predictive performance through machine learning system design.

Publications

Journal Articles

1. Y. Liu, **B. H. Lu**, J. Peng, and Z. Zhang, “Research on the Use of YOLOv5 Object Detection Algorithm in Mask Wearing Recognition,” *World Scientific Research Journal*, vol. 6, no. 11, pp. 276–284, 2020.
2. **B. Lu**, C. Moya, and G. Lin, “NSGA-PINN: A Multi-Objective Optimization Method for Physics-Informed Neural Network Training,” *Algorithms*, vol. 16, no. 4, p. 194, 2023.
3. **B. Lu**, Z. P. Hao, C. Moya, and G. Lin, “FPINN-DeepONet: A Physics-Informed Operator Learning Framework for Multi-Term Time-Fractional Mixed Diffusion-Wave Equations,” *Journal of Computational Physics*, vol. 538, p. 114184, 2025.
4. **B. Lu**, C. Mou, and G. Lin, “iPINN: An Iterative Physics-Informed Neural Network with Ensemble Kalman Filter,” *Journal of Computational Physics*, p. 114592, 2025.

Preprints and Working Papers

5. **B. Lu**, J. Zhang, and G. Lin, “Muon with Spectral Guidance: Efficient Optimization for Scientific Machine Learning,” *arXiv preprint arXiv:2602.16167*, 2026.
6. C. Mou, **B. Lu**, and G. Lin, “Neural-POD: A Plug-and-Play Neural Operator Framework for Infinite-Dimensional Functional Nonlinear Proper Orthogonal Decomposition,” *arXiv preprint arXiv:2602.15632*, 2026.
7. **B. Lu**, C. Mou, and G. Lin, “Morephy-Net: An Evolutionary Multi-objective Optimization for Replica-Exchange-based Physics-informed Neural Operator Learning Networks,” *SSRN*, 2025.
8. **B. Lu**, R. Zhang, C. Mou, N. Li, and G. Lin, “AdamFLIP: Adaptive Momentum Feedback Linearization Optimization for Hard Constrained PINN Training,” *arXiv preprint arXiv:2605.08408*, 2026.
9. **B. Lu**, Z. Deng, R. Zhang, B. Hu, Y. Zhao, Y. Tian, C. Mou, G. Lin, and X. Li, “Muon-OGD: Muon-based Spectral Orthogonal Gradient Projection for LLM Continual Learning,” *arXiv preprint arXiv:2605.08949*, 2026.
10. X. Li, J. Hou, Z. Deng, Z. Zhang, T. Li, **B. Lu**, B. Hu, Y. Zhao, and Y. Hao, “Chain of Risk: Safety Failures in Large Reasoning Models and Mitigation via Adaptive Multi-Principle Steering,” *arXiv preprint arXiv:2605.05678*, 2026.

Professional Service

Reviewer for

- *NeurIPS 2026*
- *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*
- *IEEE Data Science and Advanced Analytics (DSAA)*
- *SIAM Multiscale Modeling and Simulation (MMS)*
- *Journal of Computing and Information Science in Engineering*

Selected Research Projects

Continual Learning for Large Language Models

- Studied optimization methods for mitigating catastrophic forgetting in sequential fine-tuning of language models.
- Explored spectral-guided and projection-based update rules for stable continual learning.

Physics-Informed Scientific Machine Learning

- Developed PINN and neural operator methods for PDE learning, inverse problems, and time-fractional systems.
- Investigated multi-objective, ensemble, and evolutionary optimization methods for robust scientific ML training.

Technical Skills

Programming: Python, C/C++, Java, SQL

Frameworks/Libraries: PyTorch, TensorFlow, NumPy, SciPy, Scikit-learn

Tools: Git, Linux, Docker, MongoDB, Azure ML, HPC/Slurm

Research Areas: PINNs, Neural Operators, Scientific Machine Learning, Generative AI, LLMs, Continual Learning, Optimization